

Chronos

Daily Work Summary - 23 August 2026

Today converted Chronos from a storage-engine prototype into a coherent final-submission platform: a durable backend API, an integrated metrics-first frontend, a reproducible benchmark path, and a paper-ready research draft. The core remains the Python fixed-depth Merkle hash trie with incremental hashing and Chronos-H adaptive diff selection.

40	2	14	5
Python tests passed	Rendered frontend tests passed	Research-draft PDF pages	Benchmark variants specified

Work sequence

1	Backend API	Durable repository workflows exposed and tested.
2	Integrated frontend	Repository, import, commit, history, checkout, diff, and metrics UI.
3	Visual refinement	Professional project styling, then a white metrics-first research layout.
4	Paper draft	Architecture, methodology, literature matrix, preliminary CSV evidence, and completion gate.
5	Handoff	Verified PDFs, commit notes, reproducible generators, and exact Git staging plan.

Current position
The demonstrable platform and paper structure are ready. The only major evidence gap is the final paper-profile benchmark with Dolt installed; no state-of-the-art performance claim has been made prematurely.

1. Backend and platform delivery

Repository API	Added server-controlled create, list, open, and integrity verification. Repository names resolve beneath one server-owned root rather than accepting arbitrary client paths.	Committed as d3c9593.
Data ingest	Added keyed JSON state, JSON row, and raw UTF-8 CSV import paths.	Integrated through API and frontend.
Version writes	Added atomic puts/deletes with optimistic base-version conflict detection and per-repository write locking.	SQLite remains the durable source of truth.
History operations	Exposed history, paginated checkout, structured comparison, construction metrics, storage size, and log/Merkle work metrics.	Covers every final UI action.
HTTP controls	Added dependency-free threaded serving, loopback default, CORS allowlist, structured errors, request-size limits, and OpenAPI output.	Auth/TLS/rate limits remain out of local-demo scope.

2. Frontend delivery and redesign

Integrated workspace	Built the React/Vinext repository workspace and typed Chronos API client for create/open, import, commit, history, checkout, compare, and metrics.	Committed as 3f55bd6.
Interaction states	Added loading, offline, empty, success, and error states; persistent API URL settings; visible keyboard focus; and reduced-motion behavior.	Accessibility behavior preserved through redesigns.
Professional pass	Reduced promotional styling, tightened hierarchy, simplified labels, and introduced restrained project-oriented controls.	Committed as 682b13a.
Metrics-first layout	Finalized an all-white research dashboard with high-contrast typography, thin rules, square metric cards, cobalt accents, and a compact top navigation.	Committed as 688b523.
Scale selector	Added selectable 10, 100, 10K, 100K, and 1M record targets while explicitly separating requested scale from measured repository results.	No fabricated benchmark values shown.

3. Research, experiments, and documentation

Experimental position

- Kept the primary comparison focused on Snapshot, Log-only, Chronos-H, and Dolt; Chronos-M remains a forced-Merkle ablation.
- Defined deterministic shared workloads, disjoint calibration and evaluation phases, fixed seeds, warmups, repeated trials, medians, interquartile ranges, correctness checks, and raw CSV evidence.
- Retained the latest 1,000-row smoke run only as pipeline validation. All four available Python variants were correct; Dolt was unavailable.
- Preserved the 128-operation selector threshold as a smoke calibration result, not a proven universal crossover.

Paper-ready working draft

Architecture	Documented the content-addressed object model, b=8/d=4 fixed trie, incremental copy-on-write path rewriting, adaptive diff workflow, SQLite transaction boundary, API layer, and current scope.	Includes two architecture/workflow diagrams.
Literature review	Built a nine-source matrix spanning Merkle trees, Git, Dolt, DataHub, Decibel, OrpheusDB, dataset versioning, Noms, and ForkBase.	Primary/official sources cited.
Methodology	Specified variants, workload grid, calibration policy, timing contracts, storage/memory/work measures, correctness oracle, repetition protocol, and fairness rules.	Ready for final experiment data.
Preliminary results	Generated the smoke result table and chart directly from output/benchmarks/check-again/summary.csv.	Clearly labeled non-final; Dolt absent.
Final sections	Added evidence-gated discussion, limitations, provisional conclusion, references, and reproducible benchmark commands.	Final claims intentionally deferred.

Documents and reproducibility

- Generated an editable 14-page DOCX and a matching tagged PDF for the research draft.
- Kept a reproducible Python builder for the paper and a dated Markdown commit record.
- Previously generated the platform changes summary and system architecture handoff PDFs.
- Stored raw benchmark CSV outputs outside Git according to the repository ignore policy.

4. Verification, Git state, and next actions

Verification completed

Python unit/integration suite	40 tests passed
Frontend lint	Passed with zero errors
Frontend production test	Build passed; 2 rendered-page tests passed
Research DOCX tables	12 exact-width geometry checks passed
Research DOCX accessibility	Zero findings
Research PDF	14 pages rendered and visually inspected

Committed today

- d3c9593 - feat(api): expose durable Chronos workflows
- 3f55bd6 - feat(frontend): deliver final Chronos workspace
- 4b32f24 - docs(final): record platform delivery and architecture
- 682b13a - style(frontend): refine project workspace presentation
- 688b523 - feat(frontend): redesign as white metrics workspace

Ready for the next commit

Recommended scope: research paper and daily handoff

Stage the research-paper builder, paper DOCX/PDF, its Markdown record, this daily-summary builder, and this PDF.

Do not use git add . because the worktree also contains unrelated PDFs and temporary Word lock files.

Remaining final-submission work

- Install and pin Dolt; verify its adapter and historical correctness path.
- Extend threshold calibration beyond 128 operations and freeze the selected value.
- Run the paper profile with two warmups and seven measured trials on an idle machine.
- Review the raw CSV bundle, run the trie parameter sensitivity sweep, and optionally add the declared 1M-row stress extension.
- Replace preliminary paper results, then finalize discussion, abstract, conclusion, and the submission package.